

Cryptography Homework 2

Piter Zac. Garcia B.
Vasudev Prasad Bethamcherla

Due on Monday, 6/17 10:00AM.

1. (From Stinson): The cipher-text RUPOTENTOIFV was encrypted by a Hill cipher. The plaintext is breathtaking.

Find the key (the encryption matrix M). Give the key and a clearly written description of the steps you followed to find the key. z

For Hill cipher cryptanalysis of this question we have to find the key matrix. Hill cipher is in the form of

$$Y_{l \times n} = X_{l \times n} * M_{n \times n}$$

Where Y is the cipher text, X is the plain text and M is the key. We used brute force as the plain text was small. To find M we can do the following

$$X^{-1}_{n \times n} * Y_{n \times n} = M_{n \times n}$$

X = B R E A T H T A K I N G

Y = R U P O T E N T O I F V and n ranges from 2 to 4 . We select two characters at a time from X to get a single value of Y with that we create a n x n matrix of X and Y such as

$$X = \begin{bmatrix} B & R \\ E & A \end{bmatrix}_{2 \times 2} \quad \text{and} \quad Y = \begin{bmatrix} R & U \\ P & O \end{bmatrix}_{2 \times 2}$$

$$X = \begin{bmatrix} B & R & E \\ A & T & H \\ T & A & K \end{bmatrix}_{3 \times 3} \quad \text{and} \quad Y = \begin{bmatrix} R & U & P \\ O & T & E \\ N & T & O \end{bmatrix}_{3 \times 3}$$

We find the inverse of X using Matlab function inv() and multiply X and Y to get the possible M. After finding M for 2 and three we check if multiplying X with the resultant M gives us Y. For this method it is necessary to have multiple values (n^2) of X and Y to solve the algebraic expressions for eg. we consider $n = 2$ we need 4 values of X and Y and similarly when $n=3$ we need 9 values of X and Y. Since there are only 12 characters instead of 16 we could not find the M for $n = 4, 5, 6, \dots$

Output:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

B R E A T H T A K I N G
1 17 4 0 19 7 19 0 10 8 13 6

R U P O T E N T O I F V
17 20 15 14 19 4 12 19 14 8 5 21

Source Code:

```
x=[1 17 4 0 19 7 19 0 10 8 13 6];
y=[17 20 15 14 19 4 13 19 14 8 5 21];
m=[2 4;5 1];
z=[0 0 0 0 0 0 0 0 0 0 0 0];

z2=inv(x2)*y2
z3=inv(x3)*y3
letters(x2*z2);
letters(x3*z3);

% Doubtful about it.
x4=[1 17 4 0;19 7 19 0;10 8 13 6;3 2 3 3];
y4=[17 20 15 14;19 4 13 19;14 8 5 21;15
16 13 15];

letters(b1*z4);
letters(b2*z4);
letters(b3*z4);

c1=[1 17];
c2=[4 0];
c3=[19 7];
c4=[19 0];
c5=[10 8];
c6=[13 6];

function [ch]=letters(z)
    ch=char(mod(z,26)+65)
end

x2=[1 17;4 0]
y2=[17 20;15 14];
x3=[1 17 4; 0 19 7;19 0 10];
y3=[17 20 15; 14 19 4;13 19 14];

z4=inv(x4)*y4
letters(x4*z4);

b1=[1 17 4 0];
b2=[19 7 19 0];
b3=[10 8 13 6];
a1=[1 17 4];
a2=[0 19 7];
a3=[19 0 10];
a4=[8 13 6];

% Best answer
letters(a1*z3);
letters(a2*z3);
letters(a3*z3);
letters(a4*z3);

letters(c1*z2);
letters(c2*z2);
letters(c3*z2);
letters(c4*z2);
letters(c5*z2);
letters(c6*z2);
```

2. Show the correctness of the Euclidean algorithm.

Using the version from the slides, let $a = r_k$, $b = r_{k-1}$. Also r , s , t are the computed integers, where $r = \gcd(a, b)$ and $sa + tb = r$. Now, to show the correctness of this algorithm we need to prove that the algorithm terminates, and returns (x, y) such that $ax + by$ divides both “ a ” and b for all k in the following two steps:

First - Prove that the $\gcd(r_k, r_{k+1}) \mid \gcd(r_{k+1}, r_{k+2})$ for any k .

Taking into account that $r = \gcd(a, b)$, we will compute x and y such that $r = ax + by$.

This can be easily followed by induction in the following manner:

$$r_k = x_k a + y_k b$$

This is true for $k=1$ and $k=2$ with the (x_k, y_k) pairs $(1,0)$ and $(0,1)$ respectively.

Therefore, we will proceed to prove that it holds for $k+2$.

$$\begin{aligned} r_{k+2} &= r_k - q_{k+2} * r_{k+1} = (ax_k + by_k) - q_{k+2} (ax_{k+1} + by_{k+1}) \\ &= a(x_k - q_{k+2}x_{k+1}) + b(y_k - q_{k+2}y_{k+1}). \end{aligned}$$

The above calculation proves the following identity:

$$x_{k+2} = (x_k - q_{k+2}x_{k+1})$$

$$y_{k+2} = (y_k - q_{k+2}y_{k+1})$$

If r_m is the last remainder in the sequence then, $r = \gcd(a, b) = r_m = x_m a + y_m b$

Second - Prove using induction that the last non-zero remainder r_m will divide r_i for i from 1 to m .

Base Case: Let $i=1$, then $a = 2$ and $b = 1$ and the claim holds. Let $i = 2$, then $b \geq 2$, so $a \geq 3$. So $\gcd(3, 2) = \gcd(2, 1) = \gcd(1, 0) = 1$.

Induction Step: Assume that the smallest $a > b$ for i steps are r_{i+1} and r_{i+2} , respectively. For $i + 1$ steps, let $a > b$ be the smallest integers requiring $i + 1$ steps.

Taking into account the first step of Euclid's algorithm, we have $\gcd(a, b) = \gcd(b, r)$ where $a = qb + r$ and we know that $\gcd(b, r)$ takes i steps. Since “ a ” and “ b ” are smallest, we must have $q=1$, so $a = b + r$. Again, since “ a ” is smallest, “ b ” and “ r ” must be smallest integers that require i steps in Euclid's algorithm. By induction, $b = r_{i+2}$ and $r = r_{i+1}$, thus $a = r_{i+2} + r_{i+1} = r_{i+3}$. This concludes the proof.

On the other hand, since the ideal generated by the values of the variables a and b is invariant of the loop. Suppose that the variable a in the above proof contains the value r_m when the algorithm terminates, then $\langle a, b \rangle = \langle r_m, 0 \rangle$. It follows that a and b are multiples of r_m , hence r_m is a common divisor of a and b . Therefore, r_m can be expressed in the form $r_m = ax + by$ which allows both a and b to share a common divisor that divides r , hence $r = \gcd(a, b)$. This proves that the algorithm is correct.

3. Alice sends a message to Bob using one of the following cryptosystems. Alice is bored so she sends a message consisting of the letter a repeated several hundred times. Eve knows what system is being used but not the key and intercepts the ciphertext.

For each system, state how Eve will recognize that the plaintext is one repeating letter, and decide whether or not Eve can deduce the letter and the key.

1. Shift cipher

Having chosen a+ as the plaintext allows the cipher-text to easily yield the key. For example, if all letters in the alphabet of the plaintext are only a's, then all letters in the cipher-text will be the inverse of all key letters. Therefore, Eve will be able to deduce the letter and key because of the following reasons.

- First, all letters in the cipher-text will be the same. Having said that, the “key” of the plaintext would be found by finding the inverse of the first letter of the plaintext.
- Second, since all letters of the cipher-text are the same letter, it would be easy to assume that the letters of the plaintext are all one same letter. Therefore, finding that letter would be as easy as subtracting the key from the cipher-text m.

2. Affine cipher

Having chosen the plaintext to be all a's would yield the equation $0 + B = B$ for all characters in the cipher-text. This means, all letters in the cipher-text will be the same character (beta).

- First, since all letters in the cipher-text are the same as beta, it would be easy to figure out all the letters in the plaintext are a's because “a” is the only letter on the English alphabet whose cipher character equals beta.
- Second, knowing that all letters in the plaintext are all a's would also yield the “key” since $x(\alpha - \beta) = 0(0 - \beta) = 0$.

3. Hill cipher (with a two by two matrix). For this part it matters that the repeated letter is a.

First, due to the fact that $a=0$, the cipher-text will same as the plaintext (all a's). Therefore, this allows us to know that the plaintext is composed of only a's.

Second, the “key” of this cipher-text would not be possible to compute due to the fact that we would be dealing with only 0's matrices.

2. Exercise 6.8.17 (page 194): Suppose Alice uses RSA as follows. She starts with a message consisting of letters and assigns $a=1$, $b=2$, and so on. She then encrypts each letter separately. For example, $c=3$ encrypts to $3^e \pmod n$.

Explain how Eve can find the message without factoring n .

In particular, let $n=8881$, $e=13$.

Find the message encrypting to 4461, 794, 2015, 2015, 3603.

To find the message without factoring n we went through the following steps:

- First, create a method to find $\phi(n)$. This method counts all numbers that share 1 as a common factor.

- Second, create a method to find p and q named $pq(n, \phi(n))$. This method uses the quadratic formula $p, q = \frac{(n - \phi(n) + 1) \pm \sqrt{(n - \phi(n) + 1)^2 - 4n}}{2}$ to obtain the value of p and q accordingly.

- Third, create a method $\text{find}(\phi(n), e)$ to find the exponent for decryption. This method uses the assumption that $e \cdot d \pmod{\phi(n)} = 1 \pmod{\phi(n)}$. Note that this step was not necessary to figure out the message encrypted.

- Finally, all letter from a-z where accordingly encrypted to guess the letter encrypted by comparing all letters numerical values with the given values of the encrypted letters.

Output:

For $n = 8881$

Measure of Integers that Share the Factor of 1: 8692

$N = PQ = 8881$

$Q = 83$

$P = 107$

$d = 5349$

Encrypted Message is:

$C = 2015$

$C = 4461$

$C = 2015$

$C = 794$

$C = 3603$

Letter = A $M = 1$ $C = 1$

Letter = N $M = 14$ $C = 699$

Letter = B $M = 2$ $C = 8192$

Letter = O $M = 15$ $C = 3603$

Letter = C $M = 3$ $C = 4624$

Letter = P $M = 16$ $C = 8078$

Letter = D $M = 4$ $C = 4028$

Letter = Q $M = 17$ $C = 2825$

Letter = E $M = 5$ $C = 794$

Letter = R $M = 18$ $C = 8093$

Letter = F $M = 6$ $C = 2343$

Letter = S $M = 19$ $C = 2547$

Letter = G $M = 7$ $C = 231$

Letter = T $M = 20$ $C = 1072$

Letter = H $M = 8$ $C = 4461$

Letter = U $M = 21$ $C = 2424$

Letter = I $M = 9$ $C = 4809$

Letter = V $M = 22$ $C = 633$

Letter = J $M = 10$ $C = 3556$

Letter = W $M = 23$ $C = 413$

Letter = K $M = 11$ $C = 476$

Letter = X $M = 24$ $C = 5982$

Letter = L $M = 12$ $C = 2015$

Letter = Y $M = 25$ $C = 8766$

Letter = M $M = 13$ $C = 513$

Letter = Z $M = 26$ $C = 1783$

Reading off the output program the encrypted message is:

Message = 8 5 12 12 15 = HELLO

Source Code: For a Better Reference -> Run and See Gcd.java file

```
public static int encrypt(int e, int n){
    Integer c = 0;

    output = new LinkedList<Integer>();
    output.add(1);
    for( int i=2; i<27; i++){

        int temp = 0;
        temp = (int) Math.pow(i, 2)%n;
        //      System.out.println(temp1);

        temp = temp*i;
        temp = (int) Math.pow(temp, 2)%n;
        //      System.out.println(temp2);

        temp = (int) Math.pow(temp, 2)%n;
        //
        System.out.println(temp3);

        temp = temp*i;
        //      System.out.println(temp%n);

        output.add(temp%n);
    }
    return c;
}
```

```
static Integer mod(Integer a, Integer b){
    Integer temp = 0;
    temp = a/b;
    temp *= b;
    temp = a - temp;

    return temp;
}

public static int gcd(int a, int b){
    while( a != b){
        if( a > b)
            a -= b;
        else
            b -= a;
    }
    return a;
}
```

```
public static int phi(int n){
    int phi = 0;
    int a = 1;
    while( a != n){

        int tmp = gcd(a, n);

        if( tmp == 1)
            phi++;

        a++;
    }
    return phi;
}
```

```
public static int inverse(int alp){
    int x=0;
    alp %= 26;
    while(true){
        if((alp*x)%26 == 1%26){
            return x;
        }
        x++;
    }
}
```

```
public static int pq(Integer n, Integer phn){

    int temp = ( n - phn + 1);

    int p1= (int)((temp + Math.pow((Math.pow((n - phn + 1), 2) - 4*n), 0.5))/2);
    p = p1;

    int q1= (int)((temp - Math.pow((Math.pow((n - phn + 1), 2) - 4*n), 0.5))/2);
    q = q1;

    return p1*q1;
}
```

```

public static void main(String[] args){

    int n= 8881;
    int e = 13;
    int[] m = {4461, 794, 2015, 2015, 3603};

    int phi = phi(n);

    fillLetters();

    System.out.println("For n = " + n + " " +
        "\nMeasure of Integers that Share the Factor of 1: " + phi);

    System.out.println("N = PQ = " + pq(n, phi));
    int p = (int)getP();
    int q = (int)getQ();
    int d = findD(phi, e);

    System.out.println("P = " + p);
    System.out.println("Q = " + q);
    System.out.println("d = " + d);

    int i = -1;
    System.out.println("Encrypted Message is: ");
    while( ++i != m.length)
        System.out.println("C =: " + m[i]);

    encrypt(e, n);
    i = -1;
    while( ++i != output.size())
        System.out.println("Letter = " + letters.get(i) + " M = " + (i+1) + " C = " + output.get(i));

    System.out.println("Letter = " + gcd(3,4));
}

```

3. Exercise 6.8.7 (page 193): Naive Nelson uses RSA to receive a single ciphertext c , corresponding to the message m . His public modulus is n and his encryption key is e . He agrees to decrypt any ciphertext that someone sends him as long as it is not c . Eve sends him the ciphertext $2^e c \pmod n$.

If we have the cipher-text $c = 2^e m \pmod n$ and the public key e, n . Then, we could find m in the following manner:

$M^{ed} = m \pmod n \rightarrow M = c^{ed} \pmod n$, which in this case is:

$M = (2^e m)^{ed} \pmod n$. This is also equal to:
 $(2^{ed} * m^{ed}) \pmod n$,

And since $c^{ed} = m$ then this is equal to:
 $(2^{ed} * m) \pmod n$

Now, according to the formula $e*d \pmod n = 1 \pmod n = 1$. Therefore, since $ed = 1$ then $2^{ed} = 2$. This leaves us with the following results:
 $2m \pmod n$, according to the Euler Theorem

Explain how this allows Eve to find m .

From the calculations above, it would be easy for anyone to find m . Since we are just getting twice a letter all we need to do is divide $((2^{ed} * m) \pmod n)$ by 2 and that would yield the answer to m .

Example:

$$M = (2^{ed} * m)/2 \pmod n = m \pmod n$$